

Day-20 | Post-Quantum Security

Preparing for the Quantum Era

1. Learning Objectives

- ✓ Understand why current cryptography (RSA/ECC) is vulnerable to future quantum attacks.
- ✓ Differentiate between **Quantum Cryptography** and **Post-Quantum Security (PQC)**.
- ✓ Explain the highly urgent **"Harvest Now, Decrypt Later"** threat model.
- ✓ Learn the workflow of Quantum-Safe Migration.

2. Quick Recap

- **Quantum Cryptography (QKD)**: Secures data transmission using *quantum hardware* and the laws of physics.
- **Quantum Algorithms (Shor's)**: Designed to solve specific complex mathematical problems exponentially faster than classical computers.

Transition: Today, we look at the defensive side of software. How do we protect our current classical systems *without* requiring everyone to buy quantum hardware?

3. Why Do We Need Post-Quantum Security?

Motivation: Most of the internet's security depends on the mathematical difficulty of factoring large numbers. Quantum computers will eventually break these math problems effortlessly. We must adopt new, quantum-resistant algorithms to protect sensitive data **today** before those powerful machines are built.

Critical Sectors at Risk

- **Banking/Finance**: Long-term transaction records and identities.
- **Healthcare**: Patient medical histories and genetic databases.
- **Government/Military**: Classified intelligence and defense communications.
- **Cloud Providers**: Massive distributed storage archives.

4. Concept Expansion: Quantum vs. Post-Quantum Security

Quantum Cryptography (QKD)	Post-Quantum Cryptography (PQC)
Uses quantum mechanics (photons).	Uses new classical mathematics .
Requires specialized quantum hardware .	Runs on existing computers/laptops .
Security guaranteed by physics .	Security guaranteed by math complexity .
Secures the transmission of keys.	Protects stored data and general comms .

5. The Critical Threat: Harvest Now, Decrypt Later (HNDL)

Why upgrade now if quantum computers aren't powerful enough yet?

The Attack Strategy:

1. **Harvest:** Hackers secretly copy and download encrypted data *today* over the internet.
2. **Store:** They cannot read it, so they store it safely in massive databases.
3. **Decrypt Later:** In 5-10 years, when a powerful quantum computer is built, they use it to break the old encryption and read the stolen secrets.

Conclusion: If data needs to remain secret for 10+ years (like medical or military records), it is already at risk today.

6. Core Mathematical Representation

The Security Principle

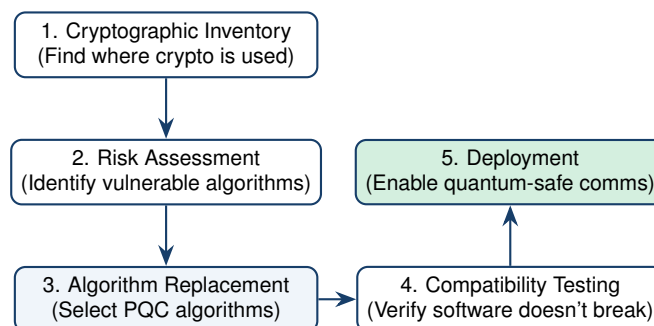
Security $\propto$ Difficulty of Mathematical Problem

Where:

- **Current Math (Vulnerable):** Factoring integers, discrete logarithms.
- **Post-Quantum Math (Safe):** Lattice-based cryptography, Hash-based signatures.
- **PQC Objective:** The algorithm must satisfy: Classical Resistant + Quantum Resistant.

7. Workflow: Quantum-Safe Migration

Organizations cannot just flip a switch to upgrade. Migration is a massive, multi-year process.



8. Deconstructing Common Misconceptions

Myth	Reality
"Post-Quantum Security requires a quantum computer to run."	No. PQC uses new classical math algorithms that run on standard laptops and servers.
"Quantum computers have already broken all encryption."	Not yet. Current NISQ hardware isn't powerful enough. We are upgrading now to prevent future "Harvest" attacks.
"Migration just means installing a quick software update."	Migration is a complex, multi-year process requiring deep network inventories and legacy system testing.

9. ★ Key Takeaways

- > **Post-Quantum Cryptography (PQC)** protects classical systems from future quantum computers using new mathematical problems (e.g., lattice-based math).
- > It is entirely software-based and runs on **existing classical hardware**.
- > The **Harvest Now, Decrypt Later (HNDL)** threat means long-term sensitive data is already vulnerable today.
- > **Quantum-Safe Migration** is the immediate global effort to inventory, test, and replace vulnerable encryption (like RSA) across the internet.